WORKSHEET 2: RAG EXPLORER

CEO of Superintelligence Workshop -- 2nd Year Edition | Time: 20 min | Individual / Pair

Part A: Concept Check (5 min)

1. RAG stands for:

R_____ A_____ G_____

2. Match the analogy:

RAG Concept	Real-Life Analogy
Storing documents in vector DB	a) Reading the chapter, then writing your answer
Searching for relevant docs	b) Putting bookmarks on important pages
Generating answer from context	c) Flipping to the right section of your textbook

Storing = ___ | Searching = ___ | Generating = ___

3. Fill in the RAG flow:

User asks a question

↓

Question is converted to _____ (numbers that capture meaning)

↓

Vector database finds the most _____ documents

↓

Relevant documents + question are sent to the _____

↓

AI answers using ONLY the _____ provided

4. What should a good RAG system do when asked about something NOT in its documents?

a) Make up a plausible answer b) Search Google instead

c) Say "I don't have that info" d) Crash with an error

Answer: ___

Part B: Design Your Knowledge Base (7 min)

Pick ONE topic (or choose your own):

- [ ] Your college/department FAQ
- [ ] A subject you're studying (DSA, DBMS, OS, etc.)
- [ ] A club or event you're part of
- [ ]

Your favorite hobby/sport rules

Your chosen topic: _____

Write 10 knowledge chunks -- one clear fact per line. Be specific!

#	Your Knowledge Chunk
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

Pro tip: More specific = better retrieval. "Library closes at 10 PM weekdays, 6 PM weekends" beats "Library has hours."

Part C: Test Your RAG (8 min)

Go to the Demo 2: RAG Pipeline slide. Paste your chunks. Load them. Then test:

3 Questions it SHOULD answer correctly:

Question	Correct?	What note did it find?
	Yes / No	
	Yes / No	
	Yes / No	

2 Questions it should NOT be able to answer:

Question	Refused?	What did it say?
	Yes / No	
	Yes / No	

Tricky question -- Can you make it hallucinate?

Your tricky question:

What happened?

Part D: Reflection

Why is RAG better than just copy-pasting everything into ChatGPT?

Name ONE real-world use case where RAG would be useful:

What happens if your knowledge chunks are vague or poorly written?

[v] Self-Check

- [ ] Explain RAG in one sentence to a friend?
- [ ] Explain what embeddings do (in simple terms)?
- [ ] Design a knowledge base for any topic?
- [ ] Predict when RAG will fail vs succeed?

You're ready to build agents! [rocket]